Micro820 v4.1.9 — Modbus Mapping Reference

Connected Components Workbench data-entry sheet. Generated 2026-05-28 from `plc/MbSrvConf_v3.xml` and `plc/live_monitor.py`.

Use: open the Micro820 project in CCW, navigate to *Controller* → *Modbus Mapping*, and enter the rows below in the order shown. Coil base = `000001` (Modbus address 0). Holding-register base = `400101` (Modbus address 100). Build, download, set to RUN.

Verify from CHARLIE: `python3 plc/live_monitor.py --host 192.168.1.100`

Currently observed from CHARLIE: TCP 502 open, every Modbus read returns exception 1 (ILLEGAL_FUNCTION) — the Modbus server map is not deployed. Loading the table below fixes it.

Discrete Coils (FC1 / FC2)

Modbus addr	CCW variable	Function / wired to	Source
000001	motor_running	Motor M-101 commanded ON	ladder
000002	conveyor_running	Conveyor CV-101 running flag	ladder
000003	fault_alarm	Composite fault latched	ladder
000004	vfd_comm_ok	Modbus poll to GS10 healthy	ladder
000005	system_ready	All preconditions met	ladder
000006	e_stop_active	E-stop circuit open	ladder
000007	dir_fwd	Selector in FWD position	input I-00
000008	dir_rev	Selector in REV position	input I-01
000009	heartbeat	1 Hz blink to prove scan	ladder
000010	estop_wiring_fault	Dual-channel E-stop XOR violation	ladder
000011	dir_fault	Both FWD and REV closed	ladder
000012	_IO_EM_DI_00	SelectorFWD (NO, knob LEFT)	input I-00
000013	_IO_EM_DI_01	SelectorREV (NO, knob RIGHT)	input I-01
000014	_IO_EM_DI_02	EStopNC (opens when pressed)	input I-02
000015	_IO_EM_DI_03	EStopNO (closes when pressed)	input I-03
000016	_IO_EM_DI_04	PBRun (illuminated momentary)	input I-04
000017	_IO_EM_DO_00	LightGreen (running indicator)	output O-00
000018	_IO_EM_DO_01	LightRed (fault / E-stop indicator)	output O-01
000019	_IO_EM_DO_02	ContactorQ1 (safety power)	output O-02
000020	_IO_EM_DO_03	PBRunLED (run pushbutton lamp)	output O-03
000021	vfd_poll_active	Modbus message in progress to VFD	ladder
000022	vfd_fault_reset_pending	Fault reset queued for next poll	ladder

Holding Registers (FC3 / FC6 / FC16)

Modbus addr	CCW variable	Type	Units / scale	Notes
400101	motor_speed	Int	RPM	Calculated from VFD freq
400102	motor_current	Int	A x 10	0..200 = 0..20.0 A
400103	temperature	Int	deg C	VFD drive heatsink
400104	pressure	Int	PSI	Spare — reserved
400105	conveyor_speed	Int	0..4095	Raw scaled VFD setpoint
400106	error_code	Int	enum	0=none 6=E-STOP 7=WIRING 8=DIR 9=VFD COMM
400107	vfd_freq	Int	Hz x 10	300 = 30.0 Hz
400108	vfd_current	Int	A x 10	GS10 actual
400109	vfd_voltage	Int	V x 10	GS10 output voltage
400110	vfd_dc_bus	Int	V x 10	GS10 DC bus voltage
400111	item_count	Int	count	Products that traversed entry sensor
400112	uptime	Int	seconds	Scan since power-on
400113	conveyor_speed_cmd	Int	0..4095	VFD speed setpoint (HMI / Modbus writes here)
400114	conv_state	Int	enum	0=IDLE 1=STARTING 2=RUNNING 3=STOPPING 4=FAULT
400115	vfd_cmd_word	Int	enum	1=STOP 18=FWD+RUN 20=REV+RUN 7=RESET
400116	vfd_freq_setpoint	Int	Hz x 10	Written from HMI / Modbus, sent to VFD reg 0x2001
400117	vfd_poll_step	Int	1..4	1=read 2=cmd 3=freq 4=fault_reset

Physical I/O Wiring (Micro820 2080-LC20-20QBB)

Terminal	CCW tag	Function	Common
I-00	_IO_EM_DI_00	SelectorFWD (NO, knob LEFT)	COM0
I-01	_IO_EM_DI_01	SelectorREV (NO, knob RIGHT)	COM0
I-02	_IO_EM_DI_02	EStopNC (opens when pressed)	COM0
I-03	_IO_EM_DI_03	EStopNO (closes when pressed)	COM0
I-04	_IO_EM_DI_04	PBRun (illuminated momentary)	COM0
I-05	_IO_EM_DI_05	Entry sensor (spare PE-101)	COM0
I-06	_IO_EM_DI_06	Exit sensor (spare PE-102)	COM0
I-07	—	(spare)	COM0
O-00	_IO_EM_DO_00	LightGreen (running)	+CM0/-CM0
O-01	_IO_EM_DO_01	LightRed (fault/E-stop)	+CM0/-CM0
O-02	_IO_EM_DO_02	ContactorQ1 (safety power)	+CM0/-CM0
O-03	_IO_EM_DO_03	PBRunLED (run pushbutton)	+CM0/-CM0

VFD Command Word — write to HR `400115`

Value	Action	GS10 reg 0x2000 meaning
1	STOP	Decel to zero
7	FAULT RESET	Clear latched VFD fault
18	FWD + RUN	Forward direction + run
20	REV + RUN	Reverse direction + run

error_code (HR 400106) decoder: 0 = none · 6 = E-STOP · 7 = WIRING fault (E-stop XOR) · 8 = DIRECTION fault (FWD and REV both closed) · 9 = VFD comm timeout.

conv_state (HR 400114) decoder: 0 = IDLE · 1 = STARTING · 2 = RUNNING · 3 = STOPPING · 4 = FAULT.

This sheet is generated; do not hand-edit. Source: tools/plc-modbus-map-pdf.py. If the ladder gains a new variable, add it to plc/MbSrvConf_v3.xml AND to this generator's tables, then re-run.